LAB 6

#1

```
#include<iostream>
#include<string>
using namespace std;

class Book
{
public:
    string title,author;
    long isbn;
};
class Library
{
    Book b[50];
    int count;

public:
    Library()
    {
        count=0;
    }
    bool addNewBook(string &title,string &author,long &isbn);
    bool removeBooks(long &isbn);
    void displayDetails();
};

bool Library::addNewBook(string &title,string &author,long
&isbn)
{
    b[count].title=title;
    b[count].author=author;
    b[count].isbn=isbn;
    count++;
    return true;
}

bool Library::removeBooks(long &isbn)
{
    for(int i=0;i<count;i++)
    {
        if(b[i].isbn==isbn)
        {
            for(int j=i;j<count-1;j++)
                b[j]=b[j+1];

            count--;
            cout<<"Removed Book";
            return true;
        }

    }
    return false;
}

void Library::displayDetails()
{
    cout<<"Details of books: "<<endl;
    for(int i=0;i<count;i++)
```

```
Enter title, author, isbn of 5 books
b1 a1 1
b2 a2 2
b3 a3 3
b4 a4 4
b5 a5 5
Now enter isbn of book to remove:
4
Removed BookDetails of books:
b1 a1 1
b2 a2 2
b3 a3 3
b5 a5 5
PS C:\Users\lenovo\Desktop\DSA>
```

```
        cout<<b[i].title<<" "<<b[i].author<<"
"<<b[i].isbn<<endl;
}

int main()
{
    Library l;
    string title,author;
    long isbn;
    cout<<"Enter title, author, isbn of 5 books"<<endl;
    for(int i=0;i<5;i++)
    {
        cin>>title>>author>>isbn;
        l.addNewBook(title,author,isbn);
    }
    cout<<"Now enter isbn of book to remove: "<<endl;
    cin>>isbn;
    l.removeBooks(isbn);
    l.displayDetails();
}
```

#2

```
#include<iostream>
#include<string>
using namespace std;

class Book
{
public:
    string title,author;
    long isbn;

    Book()
    {
        title="";
        author="";
        isbn=0;
    }

    Book(string title,string author,long isbn)
    {
        this->title=title;
        this->author=author;
        this->isbn=isbn;
    }

    Book(const Book &b)
    {
        this->title=b.title;
        this->author=b.author;
        this->isbn=b.isbn;
    }
};

class Library
{
    Book b[50];
    int count;

public:
```

```
Enter 5 books (title author isbn)
b1 a1 1
b2 a2 2
b3 a3 3
b4 a4 4
b5 a5 5
Enter ISBN to remove
4
Book List
b1 a1 1
b2 a2 2
b3 a3 3
b5 a5 5
PS C:\Users\lenovo\Desktop\DSA>
```

```
    Library()
    {
        count=0;
    }
    bool addNewBook(string title,string author,long isbn)
    {
        b[count]=Book(title,author,isbn);
        count++;
        return true;
    }

    bool removeBooks(long isbn)
    {
        for(int i=0;i<count;i++)
        {
            if(b[i].isbn==isbn)
            {
                for(int j=i;j<count-1;j++)
                    b[j]=b[j+1];
                count--;
                return true;
            }
        }
        return false;
    }
    void displayDetails()
    {
        cout<<"Book List"<<endl;
        for(int i=0;i<count;i++)
            cout<<b[i].title<<" "<<b[i].author<<"
"<<b[i].isbn<<endl;
    }
};
int main()
{
    Library l;
    string title,author;
    long isbn;
    cout<<"Enter 5 books (title author isbn)"<<endl;
    for(int i=0;i<5;i++)
    {
        cin>>title>>author>>isbn;
        l.addNewBook(title,author,isbn);
    }
    cout<<"Enter ISBN to remove"<<endl;
    cin>>isbn;
    l.removeBooks(isbn);
    l.displayDetails();
}
```

#3

```
#include<iostream>
#include<string>
using namespace std;

class Account{
    const long account_number;
    long transaction_id;
    string transaction_type;
    double balance;
```

Inputs

```
a1(1,5000);
a2(2,6000);
a3(3,7000);
a4(4,8000);
a5(5,9000);
```

```
Enter from account: 2
Enter to account: 5
Enter amount: 30
```

Output

```
public:
    Account(long acc,double bal):account_number(acc){
        balance=bal;
        transaction_id=0;
        transaction_type="none";
    }
    long depositAmount(const long to,const long from,const
double amount){
        if(account_number==to){
            balance+=amount;
            transaction_type="credit";
            transaction_id++;
            return transaction_id;
        }
        return -1;
    }
    long creditAmount(const long to,const long from,const
double amount){
        if(account_number==from){
            balance-=amount;
            transaction_type="debit";
            transaction_id++;
            return transaction_id;
        }
        return -1;
    }
    void displayDetails() const{
        cout<<"Account Number: "<<account_number<<endl;
        cout<<"Balance: "<<balance<<endl;
        cout<<"Last Transaction: "<<transaction_type<<endl;
        cout<<"Transaction ID: "<<transaction_id<<endl;
        cout<<endl;
    }
};

int main(){
    Account a1(1,5000);
    Account a2(2,6000);
    Account a3(3,7000);
    Account a4(4,8000);
    Account a5(5,9000);
    long to,from; double amount; long tid;
    cout<<"Enter from account: ";
    cin>>from;
    cout<<"Enter to account: ";
    cin>>to;
    cout<<"Enter amount: ";
    cin>>amount;
    tid=a1.creditAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a2.creditAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a3.creditAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a4.creditAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a5.creditAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;

    tid=a1.depositAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a2.depositAmount(to,from,amount); if(tid!=-1)
```

```
Transaction ID: 1

Account Details

Account Number: 1
Balance: 5000
Last Transaction: none
Transaction ID: 0

Account Number: 2
Balance: 5970
Last Transaction: debit
Transaction ID: 1

Account Number: 3
Balance: 7000
Last Transaction: none
Transaction ID: 0

Account Number: 4
Balance: 8000
Last Transaction: none
Transaction ID: 0

Account Number: 5
Balance: 9030
Last Transaction: credit
Transaction ID: 1
```

```
cout<<"Transaction ID: "<<tid<<endl;
    tid=a3.depositAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a4.depositAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;
    tid=a5.depositAmount(to,from,amount); if(tid!=-1)
cout<<"Transaction ID: "<<tid<<endl;

    cout<<endl<<"Account Details"<<endl<<endl;
    a1.displayDetails();
    a2.displayDetails();
    a3.displayDetails();
    a4.displayDetails();
    a5.displayDetails();
}
```

#4

```
#include<iostream>
using namespace std;

class B;
class A{
    int x;

public:
    void input(){
        cout<<"Enter value for class A"<<endl;
        cin>>x;
    }
    friend int add(A,B);
};

class B{
    int y;
public:
    void input(){
        cout<<"Enter value for class B"<<endl;
        cin>>y;
    }
    friend int add(A,B);
};

int add(A a,B b){
    return a.x+b.y;
}

int main(){
    A a;
    B b;

    a.input();
    b.input();

    cout<<"Sum = "<<add(a,b);
}
```

```
Enter value for class A
10
Enter value for class B
20
Sum = 30
```

#5

```cpp
#include<iostream>
using namespace std;

class Complex{
    int real,imag;

public:
    Complex(){
        real=0;
        imag=0;
    }
    Complex(int r,int i){
        real=r;
        imag=i;
    }
    Complex(const Complex &c){
        real=c.real;
        imag=c.imag;
    }
    void display(){
        cout<<real<<"+"<<imag<<"i"<<endl;
    }
    friend void sum(Complex,Complex);
};
void sum(Complex c1,Complex c2){
    Complex c;
    c.real=c1.real+c2.real;
    c.imag=c1.imag+c2.imag;
    cout<<"Sum = ";
    c.display();
}

int main(){
    int r1,i1,r2,i2;
    cout<<"Enter first complex number (real imag)"<<endl;
    cin>>r1>>i1;
    cout<<"Enter second complex number (real imag)"<<endl;
    cin>>r2>>i2;
    Complex c1(r1,i1);
    Complex c2(r2,i2);
    sum(c1,c2);
}
```

```
Enter first complex number (rea
1 2
Enter second complex number (re
3 4
Sum = 4+6i
PS C:\Users\lenovo\Desktop\DSA>
```